

Examiner reports

Q1.

Intended as an easy starter, responses to this question were surprisingly poor. Many candidates failed to appreciate the need to differentiate the given expression for velocity, assuming the constant acceleration equations to provide an appropriate method of solution.

Q2.

Diagrams in part (a) were good, but algebraic expressions produced in part (b) were poor. The use of the simple principle 'speed = distance / time' proved beyond many candidates. Many seemed unaware of what constitutes an 'equation', omitting the equals sign. There were, however, some excellent solutions to this question, showing clear and concise work.

Q3.

This question well answered by the candidates with a number gaining full marks. The candidates seemed to have been helped by the printed answer given in the first part.

Q4.

Parts (a), (b) and (c) were generally answered very well.

In part (a), a few candidates tried to substitute $t = 0$ into the position vector without differentiating.

In part (c), some candidates found the acceleration, but did not go on to find the magnitude.

Part (d) was more challenging. Quite a few candidates stated that the velocity would become zero. There were some very good answers, but also quite a few that were somewhat confused.

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Q5.

Parts (a) to (c) of this question were answered well by many of the candidates, but part (d) caused more problems. There were candidates who simply multiplied the mass and the acceleration, ignoring the resistance force. The other error was to obtain 4600 instead of 4200.

Q6.

There were many good responses to the first two parts of this question and some candidates were clearly helped by the answer given in these parts. Part (c) was more challenging and a number of candidates produced some quite confused work. The main issue was that they were using incorrect heights and/or incorrect initial velocities.

Q7.

This question was generally done well, but there was one fairly common error that was seen on quite a few scripts. The same problem arose in both parts (a) and (c), where the

candidates either did not introduce a constant of integration or did not show that it was zero. There were also a few minor errors that were produced when integrating, but these were fairly rare. A few candidates tried to use constant acceleration equations, but this approach was also quite unusual.

Q8.

Part (a) of this question was done very well and posed little difficulty for the vast majority of candidates. Part (c) was a little more demanding. Some candidates simply found the product ma giving the answer 104 N; others included three terms, but made a sign error in their equation which typically produced a final answer of 69 N.

Q9.

The candidates who knew how to tackle projectile questions with confidence did well with this question. Part (a) did not present much difficulty. However, in part (b), including the correct signs and solving the quadratic equation correctly were more challenging tasks. There were some very poor attempts at solving the quadratic equation, where candidates did not realise that the quadratic equation formula was required.

Q10.

The candidates generally did well on this question, although some did make errors when differentiating or substituting for t . Part (b) was found to be more difficult and some candidates produced very confused solutions.

Q11.

There were a fair number of good responses to part (a). The candidates who did not produce a correct solution generally either missed out the difference between the initial and final heights of the ball, or made sign errors in the setting up of their equations. A few candidates made arithmetic errors. Part (b) was also done well, with some candidates being helped by the printed answer in part (a). A few candidates used incorrect times from part (a) when trying to find the range. There were only a few correct solutions to part (c). An incomplete solution, which only found the vertical component of the velocity, was quite common among those who made some progress with the solution.

Q12.

This question was answered very well by almost all of the candidates, with many complete solutions. A small number of candidates did make minor mistakes, often of an arithmetic nature.

Q13.

Parts (a) and (b) were quite well answered; particularly part (a) where the answer was given.

In part (b) a few candidates confused the initial and final velocities. Some candidates found the position by assuming that at time t the velocity of the aeroplane would be $19\mathbf{i}+13\mathbf{j}$.

Part (c) caused a great deal of difficulty for some candidates. Many did not separate the

two components of the position vector and tried to find t by working with the whole vector. One consequence of this was that final answers for the time were given as vectors. Those candidates who formed separate equations for each component of the position vector were more successful, but some did make errors solving the equations.

Q14.

Part (a) was answered quite well, with many candidates making good use of the printed answer.

Part (b) caused more problems. Some candidates only found the vertical component of the velocity and did not bring the horizontal component into their working. As a result there were relatively few correct answers for the speed, even though quite a few candidates found the vertical component correctly. There were also quite a number of candidates you did not really know how to use the height to find a time or a velocity.

Q15.

The majority of candidates made a good attempt at this question and there were many completely correct solutions. In part (a), some of the candidates considered the three stages of the motion, but made errors when calculating one or more of these distances. In part (b), a very small number of candidates confused the numbers and gave the reciprocal of the acceleration as their answer. The majority of force diagrams in part (c) were correct, but the most common error was to omit one force.

Calculating the force in part (c) caused the most difficulties. Some candidates made a sign error in their equations, but the most common error was to simply multiply the acceleration by the mass, without taking account of the resistance force.

Q16.

This question was generally done very well. The only significant errors were associated with differentiation techniques. A few candidates did not multiply by -1 when differentiating $2e^{-t}$. A few also used some unconventional rules that seemed to be based on the product rule.

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Q17.

Part (a) of this question was done well by almost all candidates. There were a lot of good solutions for the rest of the question, but there were two problems that emerged. The first was concerning the constants of integration. Some candidates did not include these in their working and simply added a 6 to the velocity component in part (b) to get the right answer. Candidates do need to show that they introduce constants when integrating and that they do evaluate them. The other problem emerged less frequently, but some candidates did try to use constant acceleration equations to find the velocity and position vectors.

Q18.

The first three parts of this question were generally done very well. Some candidates made arithmetic or algebraic errors, but these were not common place. In part (d), a few candidates simply calculated the product of the mass and the acceleration.

Q19.

Parts (a) and (b) were done well by the vast majority of candidates. Most of the candidates who lost marks here had made minor arithmetic errors. Part (c) was a little more challenging, although there were many correct solutions. The main error made by the candidates was to use 100 m instead of $(100 - 16)$ m.

Q20.

Part (a) of this question was done well by many candidates. The main reason that candidates lost marks was because they did not differentiate correctly. For parts (b) and (c), many candidates used the position vector rather than the velocity vector as the basis for their working. This fundamental error was very common. In part (c), a number of candidates worked in degrees rather than radians, obtaining a final answer of 3600 instead of 20π .

Q21.

Many of the candidates found this question more difficult than expected. Part (a) was generally done well, but problems then emerged in part (b). Here many candidates found the magnitude of the acceleration and then used this scalar quantity in part (c). It was not uncommon to see an equation that involved a velocity that was a vector and an acceleration that was a scalar. Some of the candidates who worked through correctly stopped in part (c) and did not find the distance that had been requested.

Q22.

The candidates generally did very well with this question. Apart from a few minor errors in earlier parts, the only problems arose in part (c)(ii), where a number of candidates found the difference between 768 and 450, instead of the sum.

Q23.

The candidates had very few problems with this question, with a high proportion of completely correct solutions.

Q24.

This question was quite challenging for the candidates and there were very few good solutions, although most candidates did pick up marks from one or two parts of the question. Part (a) was done by a fair number of candidates. Errors included treating the position at $t = 10$ as a velocity and not using this position vector at all. There were some good attempts at part (b) with candidates using their acceleration from part (a). Part (c) caused problems for many candidates. It was not uncommon to see answers where the time was given as a vector. Some candidates only considered one of the components. There was clearly a number of candidates who could write down a vector equation, but could not solve it to find T . In part (c)(ii), several candidates only found the velocity and did not go on to find the speed.